

Department of Computer Applications

23MX21 – Software Engineering – Test-1

Note: Answer all the questions

COCOMO table is permitted to use

Recommended to use a pencil for drawing models.

Question (a) carries 3 Marks, (b) carries 7 Marks and (c) carries 15 Marks.

Duration: 90 Minutes

Max. Marks: 50

Q.No	Sub Division	Question	Marks	BTL
1.	a)	Software engineering is the disciplined approach to designing, developing, testing, and maintaining software systems. Write your comments on this.	3	L2
	b) i)	FinTrust Solutions, a FinTech startup, is developing a secure and scalable online banking platform for financial institutions. The system must comply with financial regulations (e.g., PCI-DSS), support real-time transactions, integrate with third-party financial services such as payment gateways and fraud detection systems, and provide a seamless user experience for both customers and bank administrators. Given the evolving regulatory landscape and the need to deliver a minimum viable product (MVP) within 6 months, selecting the right software development process model is crucial. Considering these factors, identify the most suitable software SDLC model(s) and justify your choice.	4	L4
	ii)	Comment on the following statement. “ When you are going agile without DevOps, it is like racing with a tractor instead of a car. You can do laps, but it will not move faster, and ultimately you will waste a lot of fuel without having any fun. ”	3	L2
	c) i)	For the following organic project description, generate estimates for total effort, duration, and number of people needed using the COCOMO model. Progellis Technologies is developing an AI-powered financial analytics system for investment firms. The system must process large datasets, perform complex mathematical computations, and provide real-time insights. The project consists of three major modules with the following estimated sizes: Data Processing Engine: 300 KLOC, Machine Learning Module: 50000 LOC, Visualization Dashboard: 40 KLOC Consider the following cost drivers: High - database size; very high - product complexity; very high - main storage constraint; high - execution time constraints; very high - programmer capability; very low - programming language experience. Use $a=3.2$, $b=1.05$, $c=2.5$ and $d=0.38$ Also, estimate the total project cost, considering the following salary distribution: ₹ 60,000 per analyst, ₹ 85,000 per designer, ₹ 1,20,000 per programmer, ₹ 55,000 per tester and a sum of ₹ 3,50,000 for a project manager to lead the team for the entire project.	15	L4

Q.No.	Sub Division	Question	Marks	BTL
2.	a)	Identify whether the following statements express functional or non-functional requirements for an online food delivery service i. The system shall process payments through credit card, PayPal, and digital wallets. ii. The system shall handle up to 1 million daily transactions without performance issues. iii. The system shall display restaurant menus with item descriptions and prices. iv. The system shall ensure response times do not exceed 2 seconds for any user action. v. The system shall allow customers to track their food delivery in real-time. vi. The system shall have a user-friendly interface that adapts to both mobile and desktop devices.	3	L2
	b) i)	"A Clear and well-documented requirements improve communication between stakeholders and developers". – write your comments on this.	4	L2
	ii)	During the requirement gathering for an Examination Fee Payment System, a stakeholder proposed the following requirement: <i>"If a student fails to pay the exam fee on time, the system should notify the student before restricting access to the exam."</i>	3	L5
	c) i)	Depict a neat Use Case Model for an online food delivery system where users can browse restaurants, foods, avail offers, place orders, make payments, and track deliveries. Also, write a USE CASE Story to illustrate the process of ordering food online. (or)	5 10	L5
	ii)	Depict a neat USE CASE Model for a Banking Software, highlighting key functionalities and user interactions. Write a suitable USE CASE Story to illustrate the Withdrawal of cash from the bank.	5 10	L5